LLM Foundation

Language Models: manosiga qarab turib suzlarni ketma-ketligida keyingi suzni predict qiladi
Large Language Models: dataset va parameterlar juda kup buladi

Next word prediction: umumiy matnni wordlarga yani suzlarga ajratadi.
Next token prediction: matni tokenlarga ajratadi. Tokenni word bilan farqi shundaki, token – bitta suz bulsaham uni bir nechta tokenlarga buladi.

Hozrgi tadqiqochlari technologydan uzoqlashmoqda:

10 yil oldin	Bugungi kunlarda
Modelni uzi yasasidi	GPT-4 ni API ini 1 qator code bn ishlatish
Modelni uzi uqitib train qilasidi	Faqat prompt engineer ishlatadi
Datasetni uzi tuplasidi	Tayyor datasetni ishlatish
GPU chegaralarini tushunasidi	Hardware abstracklashgan (tushnarsiz)
PhD talabasi noldan uqitar edi	Kupchilik modellarga qora quti sifatida kuradi

Qora quti: tadqiqochlar nega model ishlaydimi yoki yuqmi, qanday training bulingan va hardware qanday tasir qilishini bilmaydi?
Chunki tuliq technologyni tushunish. Raw datadan assistant gacha

Frontier models

Frontier models: hozrgi eng kuchli modellar. Juda katta companylar ularni uqitisholadi. Hamamiz bilodigon ChatGPT ni GPT-4 modelini uqitish uchun tahminan $100M ketadi.

Shunga, bizham modellar orqa tomonlaridan qanday ishlashlarini bilib turishimiz kerak. Va keyin frontier model qurish imkon bulib qolganda bizham uni qilamiz.

Small models

FLOP: NN ni hisoblash narhini ulchashning standard birligi.
Yani, $1M dollar deyilmasdan FLOP orqali ulchanadi. Bu LLM ning currency si.

※ Modelni hajmi – parameterlar soniga qarab uzgaradi

Model size	Attention vs MLP FLOPs	Hulosa
Kichik model - 100 million parameter	Tahminan teng	Attention optimization muhim
Katta model - 100 billion parameter	MLP FLOPs > Attention FLOPs	Attention optimization unchalik muhim emas

Emergence

Emergence: metriclar past turadi lekin ozgina vaqtdan keyin, metriclar birdaniga kutallinib ketadi.

Emergent ability	Expected limit	What changes
In-context learning	~10B parameter	Prompt misollari bn urganish
Chain-of-thought reasoning	~100B parameter	Kup bosqichli tasklarni yechish
Kursatmalarga rioya qilish	RLHF fine-tuningdan keyin	Murakkab topshiriqlarni bajarish
Code generation qilish	~7B+ parameters	Synthetic tugri code chiqarish

Bular LLM ning qobilyatlari

The bitter lesson

AI – insonlardan aqilli
- GPU la kup bulsa buldi emas, algorithmham yaxshi bulishi kerak.
- Yaxshi natija chiqishi uchun, GPU + Samaradorlik kerak buladi
- Samaradorlikni yaxshilash mashtabdan muhimroq

Berilgan GPU budget bilan eng yaxshi modelni qanday qurish mumkin?:

Resource	Asosiy savollar
Compute (FLOPs)	Qancha parameter, token? Qanday batch size?
Data (malumot)	Qanday sifat, filter? Qaysi domainlar aralashmasi
VRAM (memory)	Qanday aniqlik? Qaysi parallelizm strategiyasi?
Hardware	Qaysi GPU/TPU? Nechta node? Intercorrect tezligi?
Insoniy kuch	Qaysi tajribalarni utkazish? Nimaga ahimiyat berish?

Attention Mechanism & Transformer Architecture

Attention mechanism: model har bir output ishlab chiqarishda input qismlarini turli bulimlarga selective etibor qaratuvchi usul. Barcha position larga urganilgan weight bilan qaraladi.

Transformer Architecture: kup boshli self-attention va MLP qatlamlari. Barcha tokenlar parallel ishlaydi

bittasi bilan yaxshi LLM chiqmaydi. Shuning uchun Attention layers + MLP layers + Tokenizer larni birga ishlatib kuchli LLM yasaymiz.

Attention Mechanism & Transformer Architecture

C = 6 x N x D

C = FLOPs   N = parameters   D = training tokenlar

Nega aynan 6 kupaytirilyapdi?

- Forward propagation da 2 FLOP, Backward propagation da 4 FLOP. Ikkalasini birlashtirib 6 FLOP bulyapdi.

Kichkina modellar uchun: kichik tajribalar natijalarni katta modelga extrapolation qilish mumkin.

Katta modellar uchun: qimmat uqitishdan oldin scaling lawdan foydalanib natijani predict qilish

Chinchilla Law

Chinchilla law: optimal uqitish tokenlar soni parameterlar sonidan tahminan 20 baravar kup bulishi kerak.

Parameters	Optimal tokens	Amaliy mano
1B	20B token	~16 GB toza text
7B	140B token	~112 GB toza text
70B	1.4T token	~1.1 TB toza text
100B	2T token	~1.6 TB toza text

Data Sifat

Common crawl: milliardlab web-sahifalarni matn sifatida chiqaradigon notijorat tashkilot. LLM pre-trainingning eng katta manbai. Faqat kichik qismi sifatli.

Datada – spam malumotlar buladi. Reklamalar, repeating pagelar, auto-content, toxic content etc. lar buladi

Datani tozalash usullari:

1. Quality filtering
2. deduplication – duplicatelarni tashlab yuborish
3. language identification – english tilida ancha yaxshi ishlaydi.
4. toxicity classification – yomon malumotlar bilan uqitmashimiz kerak

Modelni sizeini 10 marta kupatirgandan kura datani quality sini 2 marta kupaytirish kerak.

Alignment & Supervised Fine-Tuning

Aligment: language modelni inson niyatlariga, maqsadlariga zararsiz bulishiga yunaltirish jarayoni. Yani model yomon bulmasi kerak.

Supervised Fine-Tuning (SFT): modelga qanday tugri javob berishni urgatish.
RLHF: inson tomonidan feedback olish.

Tokenization

Tokenization: textning raqamga uzgartirish jarayon.

Tokenization turlari:

Metric	Character Tokenization	Byte Tokenization	Word Tokenization	BPE Tokenization
Handles new words?	☑ Letter by letter	☑ Byte by byte	✕ Unknown	☑ Sub-words
Vocabulary size	Very small (~100 – 100k)	Extremely small (Exactly 256)	Infinite	Fixed & controlled
Sequence length	Very long	Extremely long	Short	Medium (highly balanced)
Industry status	Niche / research	Niche (megabyte model)	Outdated (Older NLP)	Modern standard

Summary

Term	Definition
Frontier models	Hozrgi kundagi eng kuchli modellar
Alignment	Modelni tugri ishlashini taminlash. Notogri yulga burmasdan
Attention	Input qismlarini turli bo'limlarga ajratish
BPE	Eng yaxghi Tokenization usuli.
Chinchilla law	Optimal token soni parameterga x20
Common crawl	Katta internetdagi LLM datalar. Datalardan ~20% foydali
Deduplication	Datadagi duplicationlarni uchirish
Language Identification	Datalarda boshqa tilni topish. Barcha holda english buladi
Toxicity Classification	Datalarda yomon malumotlar buladi. ularni filterlash kerak
Emergence	Modelda yangi qobilyat paydo bulishi
FLOP	LLM dagi modelni pullik birligi. currency
RLHF	Insondan feedback olish usuli
Scaling law	Model hajmi oshishi bilan samaradorlikning oshish qonuniyati
SFT	Modelga tugri javob berishni urgatish
Token	Matni bir nechta qismlarga ajratish.

Tokenization	Matni raqam holatga utkazish usuli
Transformer architecture	Self-attention va MLP layers. Barcha tokenlar parallel ishlaydi
Large Language Models	Juda kup parameterlar va datalar buladi
New Word Prediction	Matnni faqat suzlarga ajratish (token bilan ozgina boshqacha)